Keshav Balaji

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KeshavBalaji • keshav-balaji

Education

University of Pennsylvania — Philadelphia, PA
M.S.E. in Computer and Information Science (Cumulative GPA: 3.90) — *Expected Graduation May 2027*

- **Concentration**: Artificial Intelligence
- **Related Coursework**: Deep Learning for Time Series, Machine Learning Ecosystems (current), Multi-Modal AI (current), Trustworthy Machine Learning, Machine Learning, Internet and Web Systems, Algorithms

University of Wisconsin-Madison — Madison, WI
B.S. in Computer Science (Cumulative GPA: 3.96 — Dean's List) — *Sep 2022 – May 2025*

- **Related Coursework**: Deep Learning for Computer Vision, Artificial Intelligence, Operating Systems, Database Management Systems, Medical Image Analysis, Data Structures and Algorithms, Machine Organization, Discrete Math, User Interfaces, Human Computer Interaction, Matrix and Linear Algebra, Applied Statistics for Engineers

Publications

Shivashriganesh P. Mahato*, **Keshav Balaji***, Divyansha Lachi, Eva L. Dyer. "Population-Aware Contrastive Channel Embeddings for Intracranial EEG". *Machine Learning for Health (2026).*

Alexandre Andre*, Shivashriganesh P. Mahato*, Vinam Arora, **Keshav Balaji**, ..., Matthew R. Whiteway†, Liam Paninski†, Eva L. Dyer†. "BrainWideBench: Benchmarking large-scale pretraining and across-animal transfer in multi-region neural recordings". arXiv preprint, 2026. In review at *ICLR (2027).*

Vinam Arora, Divyansha Lachi, Shivashriganesh P. Mahato, Alexandre Andre, Jacob Poschl, **Keshav Balaji**, Gaelle Chapuis, Matthew R. Whiteway, Eva L. Dyer. "Pretrained Neuron-Level Representations Enable Across-Subject Zero-Shot Transfer of Neural Decoders". In review at *ICLR (2027).*

* Equal contribution. † Co-supervision.

Research Experience

UPenn Neural Data Science (NerDS) Lab — Philadelphia, PA
Graduate ML Researcher — PI: Eva L. Dyer — *Aug 2025 – Present*

- **Population-Aware iEEG Embeddings (iCLR)**: Developed a contrastive learning framework for intracranial EEG to study how population context among simultaneously recorded electrodes improves anatomical localization. Evaluated across three temporal encoders, with population-aware embeddings consistently improving zero-shot, cross-patient localization over local temporal features alone (ML4H 2026).
- **BrainWideBench**: Studied the effect of model scale, input representation, and trial structure on zero-shot brain region identification for a new neuro-foundation model benchmark on the IBL Brainwide Map. Built baselines spanning interspike-interval features (ISI), contrastive pretraining (NEMO), and end-to-end supervision on region labels (LOLCAT). Scaled and adapted models to the benchmark's data scale and variable-length trials for equal comparison, outperforming each model's original configuration (in review at ICLR 2027).
- **Zero-Shot Neural Decoding (ZEN)**: Analyzed how the choice of pretrained neuron-level encoder shapes zero-shot behavioral decoding by adapting an additional contrastive encoder. Evaluated how efficiently ZEN adapts with limited labeled data by implementing self-supervised pretrained decoder baselines, with ZEN outperforming them using only a quarter of the labeled data (in review at ICLR 2027).
- **Feature Representation Engineering**: Investigated what inductive biases different time-frequency features introduce to iEEG representation learning, comparing transforms (STFT, CWT), frequency binning, and normalization schemes for how well each retains the spectral structure that distinguishes brain regions.
- **Real-World Navigation and Gaze Decoding**: Studied single-trial linear and nonlinear decodability of path progression and gaze position/velocity from theta dynamics during real-world navigation (Seeber et al., 2025). Compared decoders across theta-filtered and broadband iEEG and scalp EEG, recovering path progression across all five subjects.

UW Madison Computer Science Department - UW Cosmos Project Madison, WI

Undergraduate ML Researcher — PI: Shivaram Venkataraman *Mar 2023 – May 2024*

- Engineered the document layout segmentation and extraction pipeline for the foundational ingestion stage of Cosmos, an open-source semantic search engine for scientific PDFs
- Finetuned Meta's Detectron2 Faster R-CNN with LayoutParser API for PDF segmentation, with 85%+ accuracy
- Replaced legacy rule-based segmentation algorithms with the fine-tuned model in production, improving equation detection accuracy by ˜12%

Professional Experience

Meta New York City, NY

Software Engineering Intern *May 2026 – Aug 2026*

- Engineered a global, country-aware server-side ranking engine for Instagram Edits to dynamically surface diverse creative asset types based on regional popularity and recency
- Extended server-side metadata schema to support per-font search indexing and tag-based asset discovery

Meta New York City, NY

Software Engineering Intern *May 2025 – Aug 2025*

- Developed Android-based Wearable application, streamlining the preview and testing of proprietary sound and haptic formats using Meta's SDKs, as part of the Wearables Core OS team
- Customized native AOSP services and implemented firmware level changes to vibration stack to improve the performance of haptic playback, which resulted in 70% reduction in added latency

Wisconsin State Cartographer's Office Madison, WI

Software Engineering Intern *Sep 2024 – Dec 2024*

- Revamped Wisconsin Public Land Survey Records website from a primitive, unnavigable website into a modern, interactive map with up-to-date township/section and historical survey information
- Utilized ArcGIS Maps SDK, Calcite, PostgreSQL, UW Digital Collections APIs to transform the website

PTC (Onshape) Boston, MA

Software Engineering Intern *Jun 2024 – Aug 2024*

- Upgraded Onshape CAD's iOS application, revamping its model version history graph with Swift and Objective-C
- Built live API polling to update the version graph in real time with Chrome DevTools
- Enhanced part studio feature filters and resolved user-facing and internal bugs across multiple releases

Additional Projects

EEG Sleep Stage Classification Under Multi-Source Distributional Shifts Jan 2026 – May 2026

- Quantified spatiotemporal and spectral distribution shifts via Maximum Mean Discrepancy (MMD) utilizing radial basis function (RBF) and Signature kernels, alongside log-power spectral density
- Benchmarked four data normalization strategies under a Leave-One-Subject-Out (LOSO) scheme to maximize cross-subject generalization, achieving a peak macro F1 score of 0.747
- Extracted dual-channel Short-Time Fourier Transform (STFT) features paired with recording-level normalization, improving F1 scores by ˜0.16 over single-channel STFT features and by ˜0.27 over dual-channel raw baselines

Deep Learning-Driven Multiclass Classification of Lymphoma Histopathology Jan 2025 – May 2025

- Developed ResNet-50 and EfficientNet-B0 models for lymphoma classification with accuracy over 95%
- Improved model precision by ˜5-10% utilizing ML techniques such as Gaussian blurs and color-jitter augmentations

Technical Skills

Languages: Python, Java, C, SQL, Swift, Objective-C, JavaScript, PHP, Kotlin, HTML/CSS, R, Pascal
Developer Tools: Git, Docker, Linux, Xcode, Android Studio, Bash, Figma, Chrome DevTools
Frameworks/Libraries: PyTorch, React/React Native, Pandas, Scikit-learn, ArcGIS, Seaborn, Numpy, Matplotlib